

PROJECT REPORT

Economic and Strategic Assessment of the Great Nicobar Project

Balancing Development, Sustainability and National Security

Based on Insights from the Conference on:

**Sustainable Development, Community Governance and Ecology Resilience in the
Andaman and Nicobar Islands**

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Abstract

The Great Nicobar Island, situated at the southernmost edge of the Andaman and Nicobar archipelago, represents one of India's most consequential but historically underutilized strategic assets. Positioned approximately 90 nautical miles from the Malacca Strait — through which nearly a third of global maritime trade passes — the island holds both immense economic promise and critical security relevance. The NITI Aayog's 2021 holistic development plan, involving an outlay of approximately Rs. 72,000 crore, envisions transforming this remote territory into a transshipment hub, dual-use airport, township, and energy corridor.

This project undertakes a structured assessment of the Great Nicobar initiative across four dimensions: geopolitical and strategic significance, economic and trade potential, regional connectivity architecture, and environmental sustainability. Drawing from expert deliberations at a national conference featuring Admiral D.K. Joshi (Retd.) and former Governor Shri Jagdish Mukhi, alongside available policy documents and strategic literature, the study argues that the project's success depends not on choosing between development and conservation, but on designing a governance framework that genuinely integrates both. A neutral, evidence-based conclusion is offered alongside actionable policy recommendations.

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1. Introduction

The Indian Ocean has never been simply a body of water. Throughout history, it has served as the central artery of civilizational exchange, commercial traffic, and geopolitical rivalry. In the contemporary period, as global supply chains grow more fragile and maritime chokepoints acquire renewed strategic weight, island territories have emerged from the margins of policy attention to become critical nodes of national interest. Among India's most underutilized assets in this regard is the Andaman and Nicobar archipelago — a chain of over 570 islands extending nearly 800 kilometres from north to south, and at its southern extremity, the island of Great Nicobar.

The Andaman and Nicobar Islands have been described by strategic thinkers as India's equivalent of Diego Garcia — the mid-ocean base through which the United States projects power across the Indian Ocean. Like its American counterpart, the Andaman chain allows whichever power effectively controls it to monitor the surrounding sea lanes, respond to crises rapidly, and shape the strategic environment across a vast maritime expanse. Yet for much of independent India's history, the archipelago has remained a peripheral concern — lightly administered, economically underdeveloped, and strategically underutilized.

This began to change with the formation of the tri-service Andaman and Nicobar Command in 2001, India's first and only joint theatre command. The 2021 NITI Aayog proposal for the holistic development of Great Nicobar Island represented a further and far more ambitious step. The project envisions a transshipment port at Galathea Bay, a dual-use greenfield airport at Sri Vijayapuram, a township, and a power plant — together a package designed to project economic and military presence at a location of irreplaceable geographic advantage.

However, the initiative has not proceeded without controversy. Great Nicobar is home to the Shompen, one of India's most isolated indigenous communities, and supports biodiversity of global significance. Environmental clearances granted to the project have been challenged in courts and questioned by conservationists. The tension between

development ambition and ecological responsibility forms one of the central threads running through this study.

This project examines the Great Nicobar initiative through multiple lenses — strategic, economic, ecological, and regional — drawing upon expert insights shared at a national conference by Admiral D.K. Joshi (Retd.) and former Governor Shri Jagdish Mukhi, alongside available policy literature.

2. Geopolitical and Strategic Significance

2.1 Location as Destiny

Former Governor Jagdish Mukhi, reflecting on the Andaman and Nicobar Islands during the conference, offered a phrase that captures the archipelago's essential character: 'Geography is destiny, but the ocean is an identity.' For a maritime civilization such as India's, this observation carries particular weight. The Andaman and Nicobar chain lies at the intersection of the Bay of Bengal, the Andaman Sea, and the eastern reaches of the Indian Ocean — a position that places it squarely astride some of the most heavily trafficked shipping lanes on earth.

Great Nicobar's proximity to the Malacca Strait is, in this context, its defining geographic attribute. The strait, which separates the Malay Peninsula from the Indonesian island of Sumatra, serves as the primary maritime gateway between the Indian Ocean and the Pacific. Approximately 80,000 to 100,000 vessels transit the strait annually, carrying an estimated one-third of global maritime trade, including critical energy supplies bound for China, Japan, and South Korea. Indira Point — India's southernmost tip, located at the bottom of Great Nicobar — lies barely 90 nautical miles from this chokepoint.

2.2 The Diego Garcia Analogy

Admiral D.K. Joshi drew a direct comparison between Diego Garcia — the mid-ocean atoll that anchors American power projection across the Indian Ocean — and the Andaman and Nicobar archipelago's potential for India. Just as Diego Garcia enables the United States to maintain a persistent naval and air presence across a vast maritime expanse without relying on the goodwill of regional states, a properly developed Andaman and Nicobar establishment would grant India comparable autonomous reach across the eastern Indian Ocean.

This analogy underscores a fundamental strategic point: geography, once translated into infrastructure, becomes a force multiplier. India's challenge has been that it possesses the geography but has not yet built the infrastructure to leverage it fully. The Great Nicobar project represents the most serious attempt to date to close this gap.

2.3 The Malacca Dilemma and India's Opportunity

China's strategic planners have long been preoccupied with what they term the 'Malacca Dilemma' — the country's near-total dependence on the Strait of Malacca for its energy and trade flows, and the vulnerability this creates in the event of maritime conflict or blockade. This anxiety has driven China's massive investment in alternative overland corridors (the China-Pakistan Economic Corridor, the Belt and Road Initiative) and in securing port access across the Indian Ocean littoral (the so-called 'String of Pearls').

India's development of Great Nicobar does not merely serve its own interests — it also shapes the strategic calculus of regional competitors. A robust Indian naval and surveillance presence at the mouth of the Malacca Strait materially complicates Chinese strategic planning, precisely because it places Indian assets in a position to monitor or interdict the sea lanes upon which China's economy depends. This is the strategic logic — rarely stated explicitly in official documents but widely understood among security analysts — that underpins the national security dimension of the Great Nicobar project.

3. Economic Development Vision

3.1 Galathea Bay Transshipment Port

The centrepiece of the Great Nicobar development plan is the National Container Transshipment Port proposed for Galathea Bay. Transshipment ports function as intermediary nodes in global shipping networks, where cargo is transferred from large ocean-going vessels to smaller feeder ships for onward distribution. Major transshipment hubs — Singapore, Colombo, Port Klang — typically generate enormous economic activity and employment, and their strategic importance extends well beyond their direct commercial function.

India currently lacks a significant transshipment port on its own territory. The bulk of India-origin and India-destined transshipment cargo currently passes through Colombo, Singapore, and Dubai — a dependency that imposes costs on Indian exporters and importers while enriching foreign port operators. Galathea Bay, with its natural deep water and proximity to major east-west shipping lanes, offers India the opportunity to recapture this traffic and redirect the associated economic value.

The port is envisioned as complementing the existing deep-water facility at Atlanta Bay and the commercial port at Karmatoa — the latter having been used by Japan during its colonial administration of the islands before 1947. Together, these facilities could establish Great Nicobar as a genuine hub in the Indo-Pacific maritime logistics network.

3.2 Tourism and Hospitality

Admiral Joshi highlighted the tourism potential of the Andaman and Nicobar Islands in striking terms. The archipelago's natural endowments — pristine beaches, coral ecosystems, tropical forests, and the opportunity for adventure tourism including scuba diving — position it as a world-class destination. Crucially, a single island in the Andaman group is physically larger than most competing island nation-destinations such as the Maldives, Mauritius, or Bali.

The conference noted the interest of established hospitality groups, including the Radisson Group, in developing tourism infrastructure in the islands. The seaplane

service (Sagar Maru) was cited as having the potential to become the world's largest seaplane operator given the island geography, enabling connectivity between islands that roads and conventional airports cannot economically serve.

3.3 Energy and Industrial Potential

Beyond ports and tourism, the Great Nicobar plan encompasses energy and industrial dimensions. Oil India Limited has been involved in exploration activities around Sri Vijayapuram, and there is significant interest in ultra-deep-sea resource exploration in the surrounding waters. The greenfield airport at Sri Vijayapuram — designed for dual civilian and military use — is expected to serve as a catalyst for broader economic activity in the region.

The DBFOT (Design, Build, Finance, Operate, Transfer) model has been proposed as the primary framework for private sector participation in infrastructure development — a mechanism that allows the government to leverage private capital and expertise while retaining ultimate ownership of strategic assets.

Table 1: Key Economic Sectors of the Great Nicobar Project

Sector	Key Components	Expected Outcome
Transshipment Port	Galathea Bay, Atlanta Bay deep-water port	Reduce dependence on foreign ports; capture Indian Ocean trade
Tourism	Seaplane (Sagar Maru), hospitality groups, scuba diving	Rival Maldives/Bali; generate employment and forex
Energy	Oil India exploration, ultra-deep-sea resources	Diversify domestic energy sources
Aviation	Dual-use greenfield airport (Sri Vijayapuram)	Civilian and military connectivity
Digital	Chennai–Sri Vijayapuram sub-sea fiber optic	Digital connectivity, e-governance, telemedicine

4. Regional Connectivity and Trade Architecture

4.1 Galathea Bay as BIMSTEC Maritime Capital

One of the most ambitious visions articulated at the conference — and reflected in the strategic documents shared — is the potential for Galathea Bay to function as the maritime capital of BIMSTEC (the Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation), in a manner analogous to Singapore's role as ASEAN's maritime hub. This comparison is instructive: Singapore's economic dominance rests not on the size of its domestic market but on its position at the nexus of regional trade flows, supported by world-class port infrastructure and logistics services.

Galathea Bay occupies a similarly privileged geographic position relative to the Bay of Bengal region. If developed to its potential, it could serve as the connecting node linking the Bay of Bengal with ASEAN, the India-Middle East-Europe Corridor (IMEC), and beyond.

4.2 Multimodal Connectivity Projects

The conference presentations highlighted several existing and proposed connectivity frameworks that the Great Nicobar project could anchor:

- The Kaladan Multimodal Transit Transport Project connects India and Myanmar using sea, inland waterways, and road transport — a corridor that Great Nicobar could effectively serve as the maritime origin point for.
 - Access agreements for Chattogram and Mongla ports in Bangladesh, and the Protocol on Inland Water Transit and Trade, enable movement of goods through Bangladeshi territory.
 - The India-Myanmar-Thailand Trilateral Highway improves road connectivity between South Asia and Southeast Asia.
 - The Matarbari Port Development Project and Feni River Waterways Corridor enhance regional trade and inland navigation.
 - The PM Gati Shakti Master Plan provides the overarching national framework for integrating infrastructure development across transport modes.
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4.3 Maritime Security and Energy Connectivity

Beyond commercial logistics, Galathea Bay has been identified as a potential hub for Search and Rescue operations, Maritime Domain Awareness, and disaster relief — functions that would enhance India's regional standing and deepen its relationships with neighbouring states. The facility also holds promise as a distribution node for LNG, green hydrogen, and green ammonia across BIMSTEC countries, positioning India at the centre of the region's energy transition.

Table 2: Key Regional Connectivity Frameworks

Framework	Countries / Scope	Relevance to Great Nicobar
Kaladan Multimodal Project	India–Myanmar	Sea and inland waterway corridor
Chattogram/Mongla Port Access	India–Bangladesh	Alternate trade gateway
BIMSTEC Maritime Transport Cooperation	7 Bay of Bengal nations	Multilateral maritime cooperation
India–Myanmar–Thailand Highway	South Asia–Southeast Asia	Road connectivity complement
PM Gati Shakti Master Plan	India (national)	Integrated logistics backbone

5. Environmental and Social Sustainability

5.1 Scale of Development and Ecological Sensitivity

Great Nicobar Island is among the most ecologically significant territories under Indian sovereignty. The island supports dense tropical rainforests, mangrove systems, coral reefs, and nesting habitats for the leatherback sea turtle — one of the world's largest and most endangered marine reptiles. The Nicobar group of islands is part of a global biodiversity hotspot, and the surrounding waters are rich in marine life.

Proponents of the development project have pointed out that only a small fraction — less than two percent — of the total land area of the Andaman and Nicobar Islands is proposed for active development under the Great Nicobar plan. This framing is intended to suggest that the ecological footprint of the project is limited relative to the overall scale of the archipelago. Critics, however, argue that what matters is not the proportion of land affected but the specific nature of the ecosystems disturbed — many of which are irreplaceable.

5.2 The Shompen and Tribal Rights

The Shompen, a particularly isolated indigenous community who have historically avoided contact with the outside world, inhabit the interior forests of Great Nicobar. Large-scale infrastructure development in their territory raises serious questions under domestic law, international frameworks such as ILO Convention 169 (though India is not a signatory), and basic principles of indigenous rights. The government's handling of this dimension of the project has attracted criticism from human rights organisations and some legal scholars.

5.3 The Environment-Security Dilemma

Shri Jagdish Mukhi acknowledged at the conference that 'there is always a conflict between environment and security' — a formulation that captures the essential tension without resolving it. This tension is not unique to India or to this project. The construction of military installations in environmentally sensitive areas is a recurring dilemma across the world's navies and armed forces.

What distinguishes the Great Nicobar case is the scale of the proposed development and the degree of ecological sensitivity involved. The question is not whether any development should occur — most analysts accept that some level of strategic infrastructure is necessary — but whether the current plan adequately accounts for ecological limits, and whether less damaging alternatives have been sufficiently explored.

5.4 Green Infrastructure Opportunities

One avenue for reconciling development and sustainability lies in the nature of the infrastructure being built. Japan's JAICA grant for renewable energy infrastructure in the islands points toward a model in which new construction is designed from the outset to minimise ecological impact. The 'Smart Island' concept — raised during the conference — envisions an integrated approach to energy, water, waste management, and digital connectivity that reduces the environmental footprint of human settlement. If applied rigorously, such frameworks could allow the project to serve as a model for sustainable development rather than a cautionary tale.

6. National Security Dimensions

6.1 Andaman and Nicobar Command

India established the Andaman and Nicobar Command (ANC) in 2001 — the country's first integrated tri-service theatre command, combining elements of the Army, Navy, and Air Force under a single operational authority. The ANC was conceived precisely to address the strategic potential of the archipelago, and its establishment represented an acknowledgement that the islands required a dedicated military institution rather than ad hoc coordination among the services.

Despite its existence for over two decades, the ANC has operated with resources and capabilities that analysts widely regard as insufficient relative to the strategic weight of its area of responsibility. The Great Nicobar project, if fully implemented, would substantially enhance the operational capacity of the command.

6.2 Dual-Use Infrastructure

The dual-use character of the proposed greenfield airport at Sri Vijayapuram is central to the national security rationale for the project. In peacetime, the airport would serve civilian and commercial functions. In times of tension or conflict, it would provide the Indian Air Force and Navy with a forward operating base within striking distance of the Malacca Strait — a capability with no current equivalent in India's force posture.

Similarly, the proposed deep-water port at Atlanta Bay has clear military applications alongside its commercial function. The submarine optical fiber cable connecting Chennai to Sri Vijayapuram — while primarily designed to deliver digital connectivity, e-governance, and telemedicine services — also supports anti-submarine warfare surveillance and military communications.

6.3 INS Baaz and the Forward Military Presence

INS Baaz, India's southernmost naval air station located on Campbell Bay in Great Nicobar, currently serves as the primary military installation on the island. Its capabilities are, however, limited by infrastructure constraints. The new developments proposed

under the Great Nicobar project would significantly expand India's forward military presence and transform INS Baaz from an outpost into a substantive operational node.

6.4 QUAD and Collaborative Security

The strategic significance of Great Nicobar extends beyond India's bilateral security concerns to encompass multilateral frameworks. The Quadrilateral Security Dialogue (QUAD) — comprising India, the United States, Australia, and Japan — has identified maritime security, supply chain resilience, and critical infrastructure as priority areas for cooperation. India's development of the Andaman and Nicobar Islands directly aligns with these shared priorities and could serve as a platform for joint exercises, information sharing, and coordinated maritime domain awareness.

7. Maritime Power and Strategic Doctrine

A recurring theme of the conference was the transformation of maritime power from its classical form — centred on naval supremacy and control of sea lanes through warships — to its contemporary expression, which is defined as much by logistics networks, supply chain control, and digital infrastructure as by military hardware.

Table 3: Evolution of Maritime Power — Mahan's Era vs. 21st Century

Dimension	Mahan's Era (19th–Early 20th C.)	Contemporary Era (21st Century)
Core Objective	Naval supremacy	Supply chain resilience
Focus	Warships and battle fleets	Logistics networks and trade flows
Determinant of Power	Control of seas	Control of maritime commerce
Key Actors	Empires and navies	States, corporations, alliances
Strategic Assets	Battleships and naval bases	Ports, corridors, digital platforms
Nature of Conflict	Naval battles	Economic and hybrid warfare
Doctrine	Sea Power theory	Maritime geo-economics

This evolution has profound implications for the Great Nicobar project. The port, airport, and fiber optic cable are not merely civilian amenities — they are the infrastructure of 21st-century maritime power. India's competitors, particularly China through the String of Pearls, the Belt and Road Initiative, and the China-Pakistan Economic Corridor, have grasped this logic and acted on it with considerable energy. Gwadar Port, the Chinese base at Djibouti, and the systematic acquisition of port stakes across the Indian Ocean are expressions of precisely this contemporary understanding of maritime dominance.

India's Great Nicobar project can be understood, in this light, as a belated but necessary response — an attempt to translate geographic advantage into the infrastructure of strategic influence before the window of opportunity narrows further.

8. Critical Analysis and Cost-Benefit Assessment

8.1 Strategic Benefits

The strategic case for the Great Nicobar project rests on a foundation that is difficult to challenge in its essentials. India possesses a geographic asset of extraordinary value, located at the most significant maritime chokepoint in the Indo-Pacific. The project would convert this latent advantage into actual capability — naval, commercial, and diplomatic. The creation of a transshipment hub would reduce India's costly dependence on foreign ports for its own cargo. The dual-use airport would extend India's military reach. The BIMSTEC connectivity framework would deepen India's relationships with neighbours and partners.

8.2 Risks and Concerns

The concerns are equally real. Environmental clearances were granted in circumstances that critics argue did not adequately engage with the ecological evidence. The rights and welfare of the Shompen remain inadequately protected under existing frameworks. There is a broader question of whether an investment of this scale in a remote island — however strategically located — represents the optimal allocation of limited developmental resources relative to the needs of mainland India's population.

There is also a structural concern about the imbalance between investment patterns on India's eastern and western coasts. As observed at the conference, private investment is heavily concentrated on the western seaboard while government spending dominates the eastern coast. The Great Nicobar project may draw national resources without sufficiently catalysing private participation in the broader eastern maritime economy.

8.3 The Inter-Island Connectivity Gap

One underappreciated constraint on the project's success is the absence of adequate inter-island connectivity. The conference highlighted the example of Narcondam Island — an ecologically significant and strategically located island that lacks the connectivity channels that would allow it to function as part of an integrated archipelago system. India, unlike Indonesia (itself a vast chain of islands), has not developed the regulatory

and physical infrastructure to manage its island territories as an integrated system. Until this gap is addressed, individual infrastructure investments risk being less effective than their standalone merits suggest.

9. Conclusion and Policy Recommendations

9.1 A Neutral Assessment

The Great Nicobar project sits at the intersection of several legitimate but competing imperatives — strategic necessity, economic aspiration, environmental responsibility, and indigenous rights. A balanced assessment cannot endorse it uncritically, nor reject it on environmental grounds alone.

The strategic case for developing India's forward position in the eastern Indian Ocean is compelling and, in broad terms, well-founded. The geographic advantages of Great Nicobar are real, and India's failure to leverage them over decades has come at a measurable strategic cost. At the same time, the specific choices made in the current development plan — the scale of forest clearance, the treatment of the Shompen, the pace of environmental assessment — leave significant room for improvement.

The most defensible conclusion is that India should proceed with the core strategic infrastructure at Great Nicobar, while substantially revising its approach to environmental management and indigenous rights, and investing in the inter-island connectivity systems that would allow the archipelago to function as an integrated strategic whole.

9.2 Policy Recommendations

Table 4: Policy Recommendations

#	Recommendation	Rationale
1	Establish an independent ecological monitoring authority for Great Nicobar with statutory powers to halt construction if defined ecological thresholds are crossed.	Builds credibility and ensures environmental commitments are enforceable, not merely advisory.
2	Develop a comprehensive Shompen rights and welfare framework in consultation with anthropologists and legal experts before any further construction in interior forest areas.	Legal and ethical necessity; also reduces project risk from future litigation.

#	Recommendation	Rationale
3	Adopt the Smart Island design framework as a mandatory requirement for all new construction, prioritising renewable energy (building on the JAICA renewable energy grant) and zero-waste systems.	Reduces ecological footprint while establishing a replicable model for island development.
4	Accelerate inter-island connectivity investment — including the seaplane network (Sagar Maru), sub-sea fiber extension to additional islands, and the Andaman Trunk Route (NH-4) modernisation.	Individual investments are more productive when embedded in a functioning network.
5	Deepen BIMSTEC and QUAD integration through Great Nicobar, positioning the island as a platform for joint maritime exercises, Search and Rescue coordination, and disaster relief operations.	Multilateral engagement strengthens India's strategic relationships and shares the costs of maritime governance.
6	Commission an independent cost-benefit review of the investment allocation between Great Nicobar and other eastern maritime infrastructure priorities, ensuring the project is part of a coherent eastern seaboard strategy rather than a standalone initiative.	Addresses the identified imbalance between eastern and western coast investment patterns.

In closing, Great Nicobar represents both India's greatest maritime opportunity and one of its most demanding governance challenges. The island demands not merely investment but wisdom — the capacity to think across timescales longer than electoral cycles and across the full range of values at stake. As Admiral Joshi noted, the archipelago's tourism and strategic potential exceeds that of many celebrated island nations. Whether that potential is realised sustainably will say much about the maturity of India's development state.

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